Kyle Marino

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Education

University of Southern California – Los Angeles, CA

- **M.S. Computer Engineering (2024) - GPA: 4.0**
- **B.S. Electrical and Computer Engineering (2023) - GPA: 3.94**, Presidential Scholarship, Summa Cum Laude
- Relevant Coursework: Computational Intelligence and Neural Learning, Deep Learning, Internet and Cloud Computing, Computer Systems Organization, Graduate Probability Theory, Graduate Linear Algebra

Research

Directed Research Under Dr. Viktor K. Prasanna – USC **Feb 2023 – May 2024**

- First Author on **ME-ViT: A Single-Load Memory-Efficient FPGA Accelerator for Vision Transformers** (HiPC 2023 – **Best Paper Award**) [link]
- Designed a Vision Transformer accelerator for FPGAs, achieving up to a 17.89x reduction in memory bandwidth and a 2.16x improvement in throughput per DSP over SOTA designs

Computational Intelligence and Neural Learning Final Project – EE 689 **Aug 2023 – Dec 2023**

- Developed a novel adversarial attack for Llama 2 7B to force specific hallucinated token outputs
- Used reverse backpropagation to calculate input embeddings and cosine similarity for discrete token search

Experience

ML Engineer – Eta Compute **Jan 2025 – Current**

- Leveraged Eta Compute's Aptos platform to generate SOTA edge ML models for ECG classification, achieving a test accuracy of 95% across 7 heart pathologies
- Wrote embedded C++ to run CV models on NXP's Neutron accelerator and stream video to a Python GUI

FPGA Architecture Intern – Apple **May 2024 – Dec 2024**

- Developed Python driver and Verilog AXI interface for SPI DAC peripheral on MIX FPGA
- Integrated DisplayPort IP at 8k 60 Hz over CIO80 on Agilex7 FPGA with hardware testing platform
- Wrote a SERDES link training module and PRBS generator in Verilog for loopback testing at 1.5 GHz

RTL Integration Intern – AMD/Xilinx **May 2023 – Aug 2023**

- Built a distributed design rule verification system (1000+ jobs) in Python to analyze full-chip RTL

SOC Integration Intern – AMD/Xilinx **May 2022 – Aug 2022**

- Designed and integrated Python and Perl tools for automated EMIR reporting and simulation processes

Avionics Software and FPGA Engineer – USC Rocket Propulsion Lab **Sep 2019 – Jan 2023**

- Built a 13 channel data collection and processing platform on Intel MAX10 FPGA recording at 10 MB/s
- Designed a custom microprocessor and Python compiler for real-time quaternion integration of IMU data
- Wrote C++ drivers to interface with FPGA and battery management boards

Personal Projects

Pysystemtrade – Contributor to open source quant hedge fund **Jul 2024 – Current**

- Built a collection server to record live data on 400 instruments (~4k reqs/s) using asyncio and multithreading
- Wrote custom portfolio weighting algorithm using bootstrapping and Markowitz mean-variance optimization

Live GPT **Feb 2025 – Current**

- Fine-tuned GPT-4o for processing live transcriptions from Deepgram for real-time, conversational interactions
- Designed a retraining algorithm that critiques responses and generates corrected fine-tuning examples

3D Physics Engine **Nov 2019 - Dec 2019**

- Real-time physics and rendering engine in Java with custom shader and rigid-body collision GPU kernels

Other

- Programming Skills: Python, LLMs, Vision Transformers, C/C++, Verilog, Java, Javascript, SQL, Mongo DB
- Activities: Eagle Scout, Rock Climbing, Guitar, Kendama